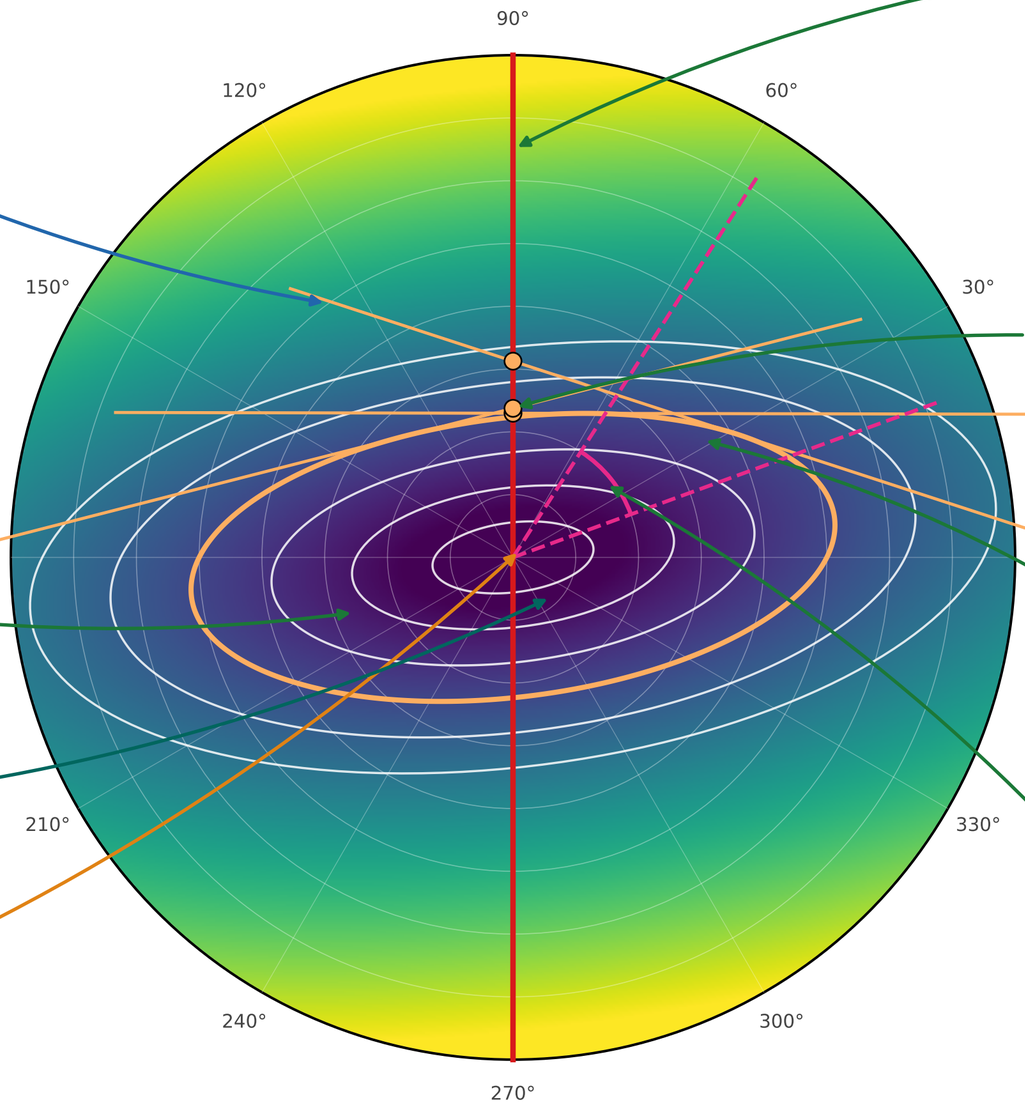
Stage 1 — Closed lobes: the ellipse phase

Norm conic deg(m − nF) = m² − a·mn + q·n² for 11a1 at p = q = 5, a = a_5 = 1 (a^2 = 1 < 4q = 20). Colour = √deg; white = level ellipses.

nested rings = weight strata |z| = q^(−σ)
= attenuation classes; purity = one ring
= distortionless (÷ q^½ → lossless)
[Restatement for Frobenius / Analogy for ζ]

y-axis = critical line Re s = ½:
the unitary spectrum of scaling
U_λ f(x) = λ^½ f(λx) on half-densities
(|λ̄^(½−s)| = 1 ∀λ ⇔ Re s = ½; TAA Lemma 153 §5)
[Theorem; frame axis = Analogy]



tangent y-intercepts = Legendre transform:
intercept = −h*(slope) carries the PHASE only;
amplitude |S″|^(−½) (half-density), Maslov
e^(iπ sgn S″/4)
(TAA Lemma 153 §5: stationary phase, rel. err 0.003 at
λ = 80)
[Theorem]

LC circuit: energy ½LI² + ½CV² is an ellipse
in the (I, V) plane iff L, C > 0 —
physical positivity, the input MP4 lacks for ζ
[Theorem for the conic; Analogy for the link]

closed lobe = definite form:
deg ≥ 0 (it counts a kernel) ⇔ a^2 < 4q
⇔ V = F̄ (FV = q) ⇔ |α| = √q (Hasse)
11a1: max|a_p|/√p = 1.947 < 2 (302 primes,
TAA Lemma 153)
[Theorem for curves]

resonance doublet at the cusp: each zero gives
two
Breit–Wigner lines at γ/2 with HWHM β/2 and
(1−β)/2
(sum ½); on the line both ¼ — one lifetime
(TAA Lemma 153 Thm 7.7; TAA Lemma 154)
[Theorem; doublet reading = Restatement]

origin ↔ point at infinity = the cusp:
constant term a_0(y) = y^s + φ(s)·y^(1−s),
|φ(½+it)| = 1 (TAA Lemma 153 NUM ≤ 2e−31)
[Analogy for the identification]

Laguerre: angle θ(ℓ_1,ℓ_2) = (1/2i)·log
CR(ℓ_1,ℓ_2; I, J),
I, J = circular points at infinity (the
absolute);
imaginary absolute = definite = rotations
(TAA Lemma 152, err 7.1e−16)
[Theorem]

Status tags: [Theorem] [Restatement] [Analogy] [Numerical] [Cited] — see legend.png. RH is not proved. No null rays at 45° (per J. W. McGreevy); lobes are elliptical conic sections, not elliptic curves.

After J. W. McGreevy's animation 'Flat Potential Oscillations' (Aug 2025, x.com/McGreevyMD/status/1951474097990279331). Re-rendered by Grok Bot from the analysed geometry; not J. W. McGreevy's code.